| Reg. No. | | | | | | | | | | | | | | | |
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**B.Tech / M.Tech Integrated DEGREE EXAMINATION, JULY 2024**
First / Second Semester

21PYB102J – SEMICONDUCTOR PHYSICS AND COMPUTATIONAL METHODS
*(For the candidates admitted from the academic year 2022-2023 onwards)*

**Note:**
(i) **Part - A** should be answered in OMR sheet within first 40 minutes and OMR sheet should be handed over to hall invigilator at the end of 40th minute.
(ii) **Part - B** and **Part - C** should be answered in answer booklet.

Time: 3 Hours | Max. Marks: 75

**PART – A (20 × 1 = 20Marks)**
Answer **ALL** Questions

| | Marks | BL | CO | PO |
|---|---|---|---|---|
| 1. Free electron moves always in _____ direction.<br>(A) Random directions (B) Positive directions<br>(C) Negative directions (D) Up and down | 1 | 1 | 1 | 1 |
| 2. Which of the following method for the band structure calculation is also known as "LCAO" approximation?<br>(A) Cellular method (B) Tight binding method<br>(C) APW method (D) Pseudo potential method | 1 | 1 | 1 | 1 |
| 3. What is the shape of E-K diagram for a free electron?<br>(A) Square (B) Circle<br>(C) Parabola (D) Straight line | 1 | 1 | 1 | 1 |
| 4. _____ explains the behaviour of metal, semiconductor and insulator.<br>(A) Classical free electron theory (B) Band theory<br>(C) Drude and Lorentz theory (D) Quantum free electron theory | 1 | 1 | 1 | 1 |
| 5. If "n" is number of electrons in the conduction band "p" is number of holes in the valence band, for an intrinsic semiconductor _____.<br>(A) $n = p$ (B) $n > p$<br>(C) $n < p$ (D) $n \neq p$ at T = ok | 1 | 2 | 2 | 1 |
| 6. The energy difference between vacuum level to the conduction band edge is termed as _____.<br>(A) Fermi energy (B) Work function<br>(C) Ionization potential (D) Electron affinity | 1 | 2 | 2 | 1 |
| 7. In LED what is the shape of domes made from plastics which is used to minimize the losses caused by bulk absorption thereby increasing the external efficiency?<br>(A) Hemispherical (B) Cylindrical<br>(C) Octahedral (D) Spherical | 1 | 1 | 2 | 1 |

8. Which one of the following is a three particle transition? 1 2 2 1

(A) Auger transition (B) Photon transition
(C) Phonon transition (D) Band to band transition

9. The Fermi's golden rule helps to identify the _____. 1 1 3 3

(A) Momentum of electrons (B) Transition rate per unit volume
(C) Velocity of light (D) Absorption energy of electron

10. The optical joint density of states is directly proportional to _____. 1 1 3 3

(A) $E^{-1/2}$ (B) E
(C) $E^{1/2}$ (D) $E^2$

11. The process of inducing radiative emission from a molecule with a help of another photon is called _____. 1 2 3 3

(A) Carrier generation (B) Stimulated emission
(C) Stimulated absorption (D) Spontaneous emission

12. The total current, through the junction of a photovoltaic cell is _____. 1 1 3 3

(A) $I_{Total} = -I_L$ (B) $I_{Total} = I_o\left(e^{qv/\beta KT} - 1\right)$

(C) $I_{Total} = I_o\left(e^{qv/\beta KT} - 1\right) + I_L$ (D) $I_{Total} = I_o\left(e^{qv/\beta KT} - 1\right) - I_L$

13. The resistivity of a bar measured through two probes 1 and 2 is 1 1 4 1

(A) $\rho = RV / l$ (B) $\rho = Rl / V$
(C) $\rho = Rwh / l$ (D) $\rho = Rl / wh$

14. _____ equation is very useful in understanding the transport properties such as electrical conductivity, thermal conductivity and thermoelectric power. 1 1 4 1

(A) Planck Einstein equation (B) Heisenberg equation
(C) Euler equation (D) Boltzmann transport equation

15. At room temperature, the voltage across the depletion layer for silicon is about _____ Volts. 1 1 4 1

(A) 0.3 – 0.4 Volts (B) 0.6 – 0.7 Volts
(C) 0.5 – 0.6 Volts (D) 0.2 – 0.3 Volts

16. What is the bandgap of GaAs semiconductor whose energy of top of valence band and bottom of conduction band comes to be 0.8 eV and 2.2 eV. 1 2 4 1

(A) $E_g = 1.1$ eV (B) $E_g = 2.2$ eV
(C) $E_g = 0.8$ eV (D) $E_g = 1.4$ eV

17. _____ density of states is independent of energy 1 1 5 1

(A) 3D (B) 0D
(C) 1D (D) 2D

18. In nanometer length scale the corresponding property is _____ dependent. 1 1 5 1

(A) Shape (B) Size
(C) Size and shape (D) Shape and volume

| | | | | | |
|---|---|---|---|---|---|
| 19. | In an armchair carbon nanotube, the geometry is _____. | 1 | 1 | 5 | 1 |
| | (A) (n, 0) (B) (n, m) | | | | |
| | (C) (n, 1) (D) (n, n) | | | | |
| 20. | _____ is the fabrication process for carbon nanotubes carried out at vacuum conditions. | 1 | 1 | 5 | 1 |
| | (A) Chemical vapour deposition (B) Arc-discharge method | | | | |
| | (C) Powder method (D) Physical vapour deposition | | | | |

## PART – B (5 × 8 = 40 Marks)

Answer **ALL** Questions

| | | Marks | BL | CO | PO |
|---|---|---|---|---|---|
| 21. a. | How do you define density of states (DoS)? Derive the expression for the DoS by considering 1/8 part of the sphere. | 2+6 | 2 | 1 | 1 |
| | **(OR)** | | | | |
| b. | Explain elaborately quantum free electron theory with any two merits and demerits. | 4+2+2 | 2 | 1 | 1 |
| 22. a. | Discuss about p-n junction under forward and reverse biasing using the suitable diagrams. | 4+4 | 3 | 2 | 1 |
| | **(OR)** | | | | |
| b. | What is continuity equation? Derive it for majority electrons and holes in a one dimensional semiconductor. | 2+6 | 4 | 2 | 1 |
| 23. a. | Derive the expression for joint density of photons in a semiconductor. | 8 | 4 | 3 | 3 |
| | **(OR)** | | | | |
| b. | Define spontaneous and stimulated emission. Obtain the relation between Einstein coefficients of spontaneous and stimulated emissions. | 2+2+4 | 2 | 3 | 3 |
| 24. a. | Explain elaborately with a neat diagram Schotky and ohmie junction. | 8 | 3 | 4 | 1 |
| | **(OR)** | | | | |
| b. | Explain about the four probe collinear method for a bulk sample and a thin sheet with necessary equations in order to find the resistivity of shapes with uniform cross section. | 4+4 | 2 | 4 | 1 |
| 25. a. | Explain elaborately the chemical vapor deposition method of fabrication of carbon nano tubes. Write any two advantages and disadvantages. | 4+2+2 | 2 | 5 | 1 |
| | **(OR)** | | | | |
| b. | Explain, the principle, construction and working of transmission electron microscopy with specimen interactions and utilization. | 2+2+4 | 2 | 5 | 1 |

## PART – C (1 × 15 = 15 Marks)

Answer **ANY ONE** Questions

| | Marks | BL | CO | PO |
|---|---|---|---|---|
| 26. Derive an expression for motion of electron in a periodic potential using Kronig-Penney model. Calculate the energy of particles at P = 0 and P = ∞, where P is the power of potential well. | 9+3+3 | 3 | 1 | 1 |
| 27. What is Hall effect? Derive an expression for the hall coefficient $R_H$ for an n-type and p-type semiconductor. Derive and determine the $R_H$ for a rectangular slab with thickness t and width w. | 2+4+4 +5 | 3 | 4 | 1 |

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